Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653104152
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 46 Question Id : 640653701049 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SPEECH TECHNOLOGY(PEN AND PAPER EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406532340319. ✔ YES

6406532340320. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653104153
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 47 Question Id : 640653701050 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 8

Question Label : Multiple Choice Question

1.1 MCQ *(Note: Please explain your choice)* 8 points

1. (2 points) In the context of Hidden Markov Models (HMMs), what is the purpose of the Forward Algorithm?

   A. Estimating the most likely sequence of hidden states given observed data

   B. Computing the likelihood of observed data given the model parameters

   C. Updating the transition probabilities based on the observed sequence

   D. Sampling from the posterior distribution of hidden states

2. (2 points) State whether the following statement is true or false: A Transformer Decoder layer is allowed to compute the hidden representation of all time steps simultaneously.

   A. True

   B. False

3. (2 points) Which of the following ideas is used by LSTM-based RNN networks to capture long range dependencies?

   A. Pooling

   B. Convolution

   C. Gates

   D. None of the above

4. (2 points) What is the main objective of speaker diarization in the field of audio signal processing?

   A. Speech recognition

   B. Speaker identification

   C. Clustering and labeling speakers in an audio recording

   D. Acoustic modeling

Options :

6406532340321. ✔ I have written answers on the answer sheets

6406532340322. ✖ Not applicable

Sub-Section Number :	3
Sub-Section Id :	640653104154
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 48 Question Id : 640653701051 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 32

Question Label : Multiple Choice Question

1.2 Subjective Questions — 32 points

1. (2 points) Explain the difference between local and global minima. Discuss the steps suggested in literature to avoid the problem of getting stuck at a local minima.
2. (5 points) Describe the terms kernel, stride, padding and dilation in CNNs with an example.
3. (5 points) Explain permutation invariant loss in end-to-end speaker diarisation.
4. (10 points) Describe all the steps involved in Transformer based Automatic Speech Recognition when you have raw WAV files and corresponding transcription. (You have to discuss everything detail from the start to end like STFT, CTC etc,)
5. (10 points) Describe the steps involved in deep learning based speaker verification and explain the X-vector based speaker verification along with the different error metrics to assess the verification system.

Options :

6406532340323. ✔ I have written answers on the answer sheets

6406532340324. ✖ Not applicable

English2

Section Id :	64065349389
Section Number :	4
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	27
Number of Questions to be attempted :	27
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0